

NCIE-016 Skeletal v0.1 → Full v1.0

Expansion Verification Report

Testing, Verification & Acceptance Specification — IN DEVELOPMENT / FOR HUMAN REVIEW OF FULL EXPANSION

1. Title Reconciliation Verification (Directive Item 3) — SOURCE-VERIFIED

Authoritative register used: NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL, dated 02 September 2026), located on the controlled document store. NCIE-016 entry located: Section 3, 'Reconciled Master Documentation Register' table, row NCIE-016. Authoritative formal title (as recorded in the register): 'Testing, Verification & Acceptance Specification'. Skeletal working title prior to this verification: 'NCIE-016 — Testing, Quality Assurance & Verification Specification'. Comparison result: SEMANTIC MATCH — DOCUMENT-CONTROL WORDING DIFFERS (the register's Primary Purpose text — 'Verification strategy, test suites, security/AI/data testing and acceptance gates' — matches this document's actual 40-chapter scope exactly: verification strategy Chapters 1-9, test suites across every upstream domain Chapters 10-28, and acceptance/sign-off governance Chapters 29-32; the wording difference is the register dropping 'Quality Assurance' and adding 'Acceptance', not a scope difference; no material discrepancy was found). Correction applied: the document-control title was set to 'NCIE-016 — Testing, Verification & Acceptance Specification' in the cover, §0.1, the Document Governance table, the Authority Chain, the running header, §40.6, core document properties and the deliverable filename, per directive item 3's wording-correction procedure (document-control only, no Human Review halt required). 40-chapter integrity: PASS (40/40). Salient Things to Note: PASS (40/40). Human Review Register reconciliation: PASS (40 total / 38 Blocking / 2 Non-Blocking / 0 duplicate / 0 orphan). Three-artifact consistency: PASS. This finding is SOURCE-VERIFIED — established directly from the text of the Master Controlled Register v2.0 document itself and the skeletal document's own Document Governance block, not inferred from naming convention or conversational context.

2. Scope Verification

NCIE-016 v1.0 is a controlled full expansion of the approved NCIE-016 Skeletal v0.1: every approved chapter purpose, Salient Things to Note, principal component, upstream dependency/authority and Human Review focus is elaborated into requirements traceability, test-level taxonomy, risk-based prioritization, environment/data control, automation architecture, code-level through cross-specification end-to-end testing across every NCIE-001–015 domain, defect governance, evidence/results modeling, execution independence, acceptance/sign-off, coverage reporting, reproducibility, governance-control verification, CI/CD continuous verification, and bounded production verification — without inventing test-management/automation/security-testing/performance/accessibility/chaos/observability products, coverage targets, pass thresholds, benchmark cutoffs, performance limits, defect severities, RPO/RTO acceptance values, flake tolerances, release-blocking thresholds, or test cadences absent source authority, and without claiming any test was actually executed, passed, validated, or accepted.

3. 40-Chapter Integrity Verification

A programmatic scan of the final document confirms exactly 40 sequential 'NCIE-016 — Chapter N Expansion' labels, N = 1..40, each present exactly once, in the approved order, with no chapter added, removed, merged, split or renamed relative to the approved Skeletal v0.1 structure.

4. Salient Things to Note Hard-Gate Verification

Per directive item 86 — continuing the discipline established after the NCIE-009 Chapter-27 omission — every chapter must retain an explicit '<N>.0 Salient Things to Note' subsection. This build enforced the requirement structurally: the chapter-rendering helper (lib.py's add_chapter function) raises a build-time assertion error if a chapter's salient content is omitted, so the document cannot be assembled with a missing subsection. A programmatic scan of the final assembled document, rerun independently rather than inferred from the build script succeeding, confirms an explicit Salient Things to Note heading present for all 40 chapters (N = 1 through 40), with zero missing.

5. Skeleton Fidelity Verification

Each of the 40 chapters was authored directly from its approved skeletal Chapter Purpose, Salient Things to Note, Principal Components, Upstream Dependencies/Authority and Human Review Focus (see the accompanying Change Register for the per-chapter mapping). No approved chapter concept was omitted, and none was conceptually inverted.

6. Status-Integrity / No-False-Status-Promotion Audit (Directive Items 20, 71)

A full-text scan for any claim that planned/specification evidence equals executed/passed/validated/accepted status found none — every chapter's status line explicitly states 'Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here', and a targeted pattern scan for phrases such as 'tests were executed', 'all tests passed', 'verification complete' or 'production-ready' returned zero hits across the entire document. The Master RTM (Table

7) and Acceptance/Sign-Off Matrix (Table 38) both explicitly read NOT RUN / Pending throughout rather than a fabricated result.

7. RTM Completeness Audit (Directive Item 72)

Every in-scope requirement source (NCIE-001 through NCIE-015) has a disposition in the Master RTM structure (Chapter 4, Table 7): mapped to a test class (per the Cross-Spec Test Catalogue, Table 2), blocked pending a named Human Review item, or covered by an explicit chapter (5-28) that states its verification approach. No in-scope requirement source was found without a disposition.

8. Security Assurance Audit (Directive Item 73)

Chapter 19's SECURITY SCAN PASS $\neq$ SECURITY ASSURANCE and PENETRATION TEST PASS $\neq$ ZERO RISK invariants are restated at the point of use and cross-referenced throughout; a full-text scan confirms no chapter represents a scan or penetration-test result as total security assurance or zero risk.

9. AI/Domain Validity Audit (Directive Item 74)

Chapters 16, 15, 18 and 21 were verified to keep model-benchmark, Agent-task, VPF and domain-scenario success distinct from domain validity, Finding, Decision or Human Approval — MODEL BENCHMARK PASS $\neq$ DOMAIN VALIDITY, AGENT TASK SUCCESS $\neq$ AUTHORITY COMPLIANCE, VPF PASS $\neq$ HUMAN APPROVAL and 'tests cannot manufacture Findings' are each restated at their owning chapter.

10. Accessibility/Usability Audit (Directive Item 75)

Chapter 22 was verified to keep automated accessibility-tool results and usability-session results distinct from accessibility conformance and representative-user acceptance — ACCESSIBILITY TOOL PASS $\neq$ ACCESSIBILITY CONFORMANCE and USABILITY SESSION $\neq$ REPRESENTATIVE-USER ACCEPTANCE are both restated, with representative-user sampling named as a deliberate Human Review decision rather than assumed.

11. Recovery/DR Audit (Directive Item 76)

Chapter 24 was verified to keep backup-test, DR-exercise and game-day results separated from production recovery proof and current-state reauthorization/reconciliation — BACKUP TEST $\neq$ RECOVERY VALIDATION and DR EXERCISE $\neq$ PRODUCTION RECOVERY PROOF are both restated, consistent with the identical invariants in NCIE-014/015.

12. Performance/Capacity Audit (Directive Item 77)

Chapter 25 was verified to scope every performance result to an explicit workload/environment rather than generalizing to unlimited capacity assurance — PERFORMANCE TEST $\neq$ CAPACITY ASSURANCE is restated, and §25.1 requires a workload/environment to be stated for every reported number.

13. Defect/RCA Audit (Directive Item 78)

Chapter 29's Defect-State Matrix (Table 33) was verified to keep defect closure, deferred/accepted risk acceptance and root-cause elimination as three distinct paths — DEFECT CLOSED $\neq$ ROOT CAUSE ELIMINATED is restated, and the matrix routes deferred/accepted defects to NCIE-008 accountable authority rather than the fix implementer alone.

14. Test-Data Privacy Audit (Directive Item 79)

Chapter 8 was verified to require synthetic/masked data by default, explicit Human Review authorization for any production-derived data, and provenance/refresh/access/cleanup tracking — TEST DATA $\neq$ PRODUCTION AUTHORITY is restated, consistent with NCIE-009 classification rules.

15. No-Fabricated-Tool Audit (Directive Item 80)

A full-text scan for unsupported test-management, automation, security-testing, performance, accessibility or chaos tooling selections found none — every tool decision (Chapters 9, 19, 22, 25-26) is stated as a Human Review item (e.g. HR16-9-1, HR16-19-1, HR16-22-1, HR16-25-1, HR16-26-1) rather than a mandated product.

16. No-Fabricated-Threshold Audit (Directive Item 81)

A full-text scan for invented coverage/pass/benchmark/performance/severity/flake/cadence/RPO-RTO/release-blocking values found none — Chapter 10 defers coverage targets (HR16-10-1), Chapter 16 defers benchmark thresholds (HR16-16-1), Chapter 24 defers RPO/RTO (HR16-24-1), Chapter 25 defers workload targets (HR16-25-1) and Chapter 33 defers quality-metric thresholds (HR16-33-1) — every concrete number is a named Human Review item, never an invented default.

17. No-Execution-Claims Audit (Directive Item 82)

This production task specified the verification programme only, per directive item 82's explicit instruction; no implementation evidence was supplied to this production run, so no test anywhere in the document is claimed executed or

passed. The programmatic pattern scan described in §6 above independently confirms zero such claims in the final assembled document.

18. Adversarial Reviewer Pass (Directive Item 83)

The document was reviewed from QA-architecture, security-testing, AI/Agent-evaluation, data/migration, UX/accessibility, resilience/performance, Evidence/audit and governance/acceptance perspectives. Corrections applied during review: ensured the Acceptance/Sign-Off Matrix (Table 38) explicitly shows a Pending decision for every illustrative row rather than an empty cell that could be misread as an implicit pass; ensured the Defect-State Matrix (Table 33) routes deferred/accepted defects through NCIE-008 accountable authority specifically, not merely 'an approver', closing a gap that could otherwise let a fix implementer self-accept residual risk. No defect exposing private chain-of-thought was found or is reported.

19. Skeleton Fidelity Audit

All 40 approved Chapter Purposes, Salient Things to Note, Principal Components, dependencies and Human Review foci were verified represented and not inverted, per the mapping in the accompanying Change Register.

20. 40-Chapter Integrity Audit (Directive Item 85)

Programmatically re-verified: exactly 40 sequential chapters, no missing/duplicate/added chapter and no unintended renaming.

21. Salient Things to Note Audit (Directive Item 86)

Programmatically re-verified: an explicit Salient Things to Note subsection is present in all 40 chapters.

22. Human Review Register Mechanical Reconciliation

Chapter 39's register was recomputed programmatically by scanning the Human Review Focus bullet text already rendered in Chapters 1-38, plus HR16-39-1 (Chapter 39's own item) and HR16-40-1 (Chapter 40's own item) — both defined once and included in the register-construction list from the start of the build, a design choice made specifically to avoid repeating the register-reconciliation defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Result: 40 total items, 38 Blocking / 2 Non-Blocking, 0 duplicate Review IDs, 0 orphan Human Review items. Classification breakdown: 27 Institutional Confirmation Required, 6 Security/Sovereignty Confirmation Required, 5 Engineering/Infrastructure Discovery Required, 1 Legal/Policy Confirmation Required, 1 Resolved by Upstream Baseline.

23. Upstream Human Review Deduplication (Directive Item 70)

Every NCIE-016 Human Review item was individually checked against the existing NCIE-008–015 Human Review registers. None was found to restate an identical authority, threshold, environment, security, UX or operations question — each concerns a distinct testing decision those upstream documents raise the need for but do not themselves resolve. No cross-reference deduplication was required for this edition.

24. Blocking Scope Discipline

Each Blocking item in the Chapter 39 register states precisely which tool selection, numeric threshold, scenario corpus or acceptance authority it blocks, rather than implying the whole document or the whole approved upstream chain is blocked. Non-Blocking items (Chapter 34, and the resolved Chapter 40 item) are explicitly marked as not gating the corresponding chapter's verification activity in the interim.

25. Master RTM, Test-State, Cross-Spec Catalogue, Defect-State, Test-Evidence and Acceptance/Sign-Off Matrix Verification (Directive Items 63-68)

Table 7 (Chapter 4) was verified to structure a Master RTM entry for representative requirement classes across NCIE-001–015 with every Result cell reading NOT RUN. Table 4 (Chapter 2) was verified to fix seven non-substitutable test states from Specified through Human/Production Acceptance. Table 2 (Chapter 1) was verified to map every upstream domain to required test classes and an unresolved-decision reference. Table 33 (Chapter 29) was verified to define nine defect states with deferred/accepted routed to NCIE-008 accountable authority. Table 35 (Chapter 30) was verified to fix minimum evidence fields across six representative test classes. Table 38 (Chapter 32) was verified to map four capability/risk classes to required evidence, test status, HR blockers and a Pending decision throughout.

26. Character-Encoding Integrity Check

Following the same build pipeline used for the whole NCIE series (UTF-8 source fragments executed via a Python exec() call with an explicit encoding='utf-8' argument, a fix first required during the NCIE-008 build), a full-document scan of the final NCIE-016 document confirmed zero mojibake occurrences across all paragraphs and tables.

27. Visual and Structural Verification

The complete document was opened in Microsoft Word, its Table of Contents field (a live Word field, not static text) was updated and confirmed to populate with correct chapter/section titles and real page numbers, repaginated to 65 pages, saved, and rendered to PDF. Independent text extraction from the rendered PDF confirms the cover occupies page 1 alone and Document Governance begins on page 2 (the manual page break), with the running header and footer correctly reading NCIE-016 branding and the reconciled title. No table clipping or broken table was observed across the six consolidated matrices, the Chapter 39 Human Review Register tables, or the Chapter 40 traceability/handover tables; the 11-column Test-Evidence Matrix (Table 35) wraps tightly within narrow columns on a portrait page but no cell content is cut off or overlapping. A targeted scan for the stale wording 'Quality Assurance & Verification Specification' used as a formal title (rather than the intentionally preserved historical working-title quote) found zero occurrences, and a targeted scan for execution-claim language ('tests were executed', 'all tests passed', 'verification complete', 'production-ready') found zero occurrences.

28. Final Human Review Extraction (Authoritative)

Performed directly against the final specification file by independently scanning every paragraph and table cell for the pattern HR16-<chapter>-<n>: TOTAL IDS FOUND = 40. UNIQUE IDS = 40. DUPLICATES = 0. OPEN = 39. RESOLVED = 1 (HR16-40-1, Resolved by Upstream Baseline). BLOCKING = 38. NON-BLOCKING = 2. CLASSIFICATION TOTALS: Institutional Confirmation Required = 27; Security/Sovereignty Confirmation Required = 6; Engineering/Infrastructure Discovery Required = 5; Legal/Policy Confirmation Required = 1; Resolved by Upstream Baseline = 1. ORPHAN IDS = 0 — every one of the 40 unique IDs found anywhere in the document also appears as a row in the Chapter 39 Complete Register table, which itself contains exactly 40 rows with no duplicate Review ID.

29. Three-Artifact Consistency Check

The NCIE-016 v1.0 main document, this Verification Report and the accompanying Full-Expansion Change Register agree exactly on: Formal title 'NCIE-016 — Testing, Verification & Acceptance Specification' (reconciled at production time per §1 above, SOURCE-VERIFIED against Master Controlled Register v2.0); Source Version v0.1 (Skeletal) → Target Version v1.0 (Full); Authority chain approved NCIE-001 v1.3 through approved NCIE-015 v1.0, plus the approved NCIE-016 Skeletal v0.1; Status IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION (not Approved); 40 chapters preserved in order (0 missing, 0 duplicate, 0 added, 0 unauthorized renaming); Salient Things to Note hard gate = 40/40 explicit subsections present; Human Review Register TOTAL = 40, BLOCKING = 38, NON-BLOCKING = 2, classification totals as in §28 above; orphan Human Review IDs = 0; duplicate Human Review IDs = 0; test-status posture = every test SPECIFIED only, none executed/passed/validated/accepted anywhere in the document; testing/verification invariants = 21 NCIE-016-specific plus cross-referenced upstream invariants; acceptance boundary = Chapter 32's matrix reads Pending throughout, with final acceptance authority resting entirely with NCIE-008 downstream of this document; title-reconciliation result = SEMANTIC MATCH / DOCUMENT-CONTROL WORDING DIFFERS, corrected. No mismatch was found between the three artifacts.

30. Final Readiness Disposition

NCIE-016 — Testing, Verification & Acceptance Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed tool selection, numeric threshold, acceptance authority and scenario-corpus assignment deferred in Chapters 1-38 remains a proposal pending resolution of the Chapter 39 Human Review Register (including the formal-title confirmation, HR16-1-1). No test described anywhere in this specification has been executed, passed, validated, or accepted — this document is a verification programme, not a verification result. This document is NOT marked Approved.